PHASE AND GAIN MARGIN

Phase margin, FM.

The phase margin is the change in open-loop phase shift required at unity gain to make the closed-loop system unstable.

Phase margin is that amount of additional phase lag at the gain cross over frequency required to bring the system to the verge of instability. Gain cross over frequency is the frequency at which $|G(j\omega)| = 1$. The phase margin γ is 180^0 plus the phase angle $< G(j\omega)$ of the open-loop transfer function at the gain cross over frequency

$$\gamma = 180^0 + < G(j\omega)$$

For a system to be stable, the phase margin should be positive.

Gain margin, GM.

The gain margin is the change in open-loop gain, expressed in decibels (dB), required at 180^0 of phase shift to make the closed-loop system unstable.

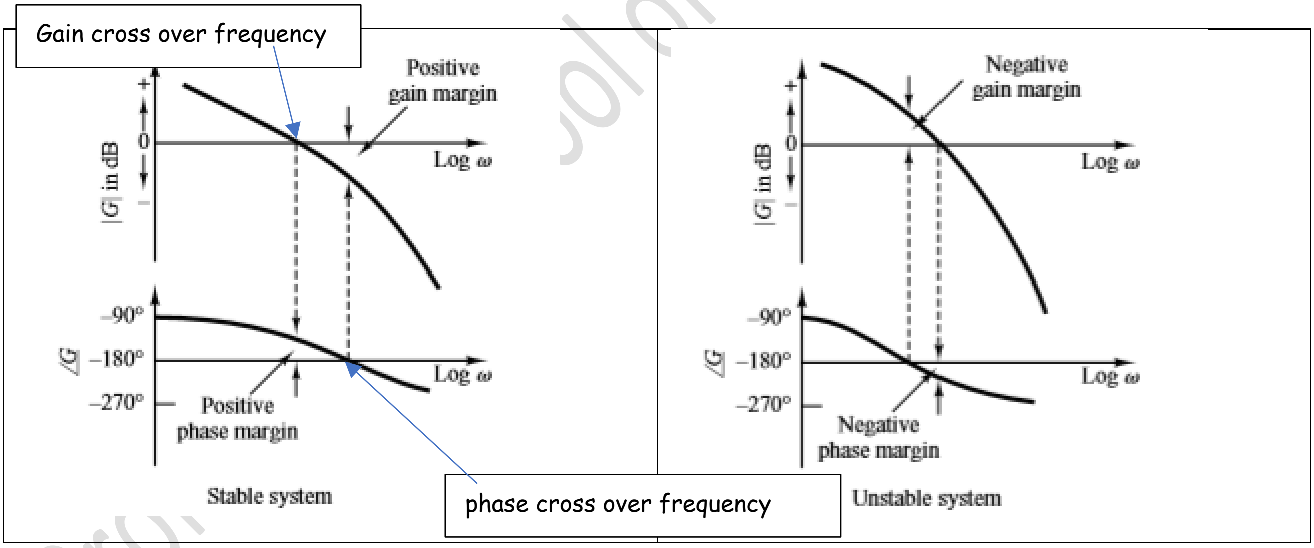
Gain margin is the reciprocal of the magnitude $|G(j\omega)|$ at the froquency at which the phase angle is -180^0

The phase cross over frequency ω_1 is defined as the frequency at which the phase angle of open loop transfer function equals -180^0.

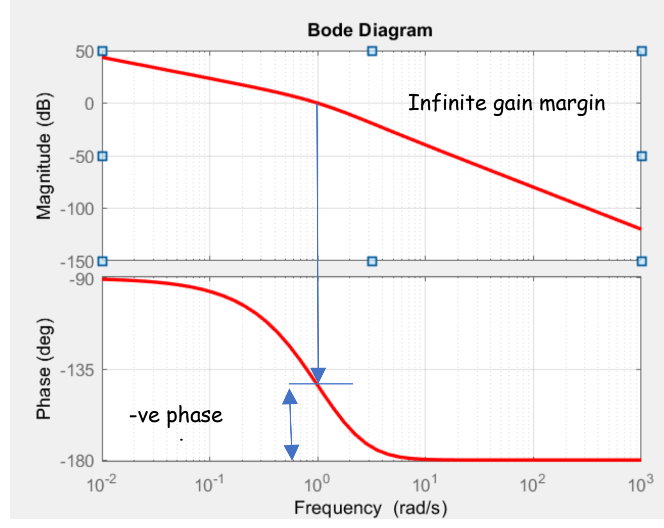
Gain margin $K_g = \frac{1}{|G(j\omega_1)|}$. In terms of decibels, $K_g\ dB = 20\log\frac{1}{|G(j\omega_1)|} = -20\log|G(j\omega_1)|$

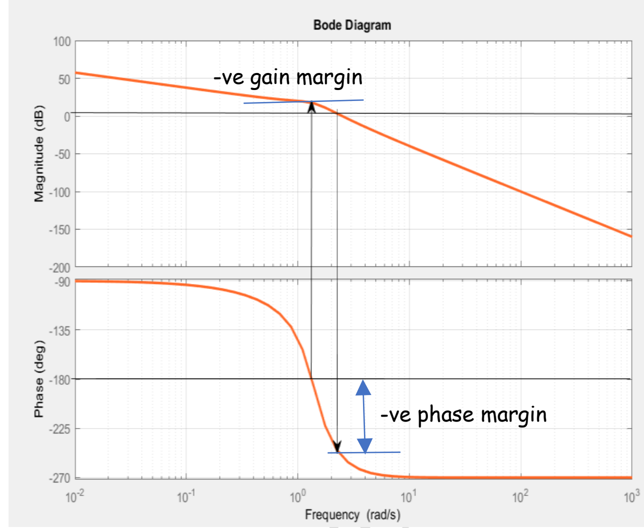
The gain margin in dB is positive if $K_g > 1$ and negative if $K_g < 1$. A positive gaian margin in dB means the system is stable.

The figure below shows the gain margin and phase margin of a stable and unstable systems.



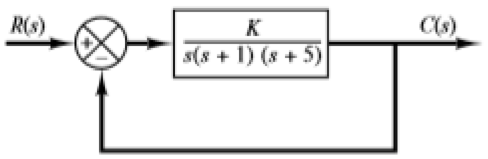
The figures below show the Bode diagram of the examples we solved above. Identify the gain and phase margins and comment on the stability of the systems.

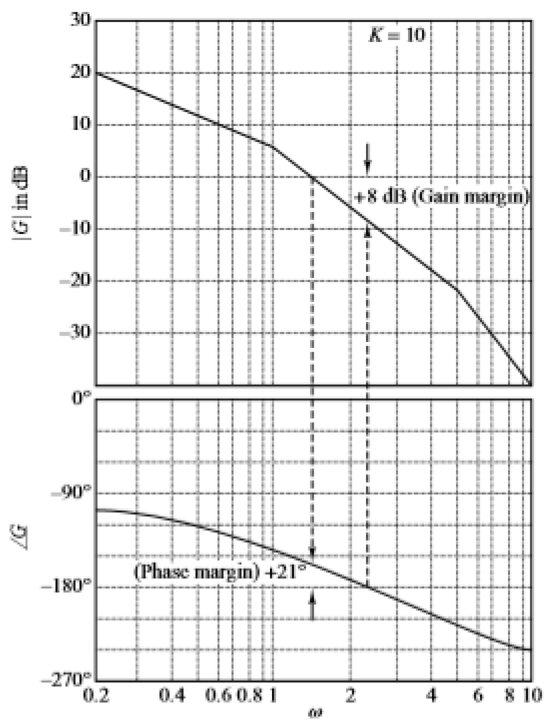


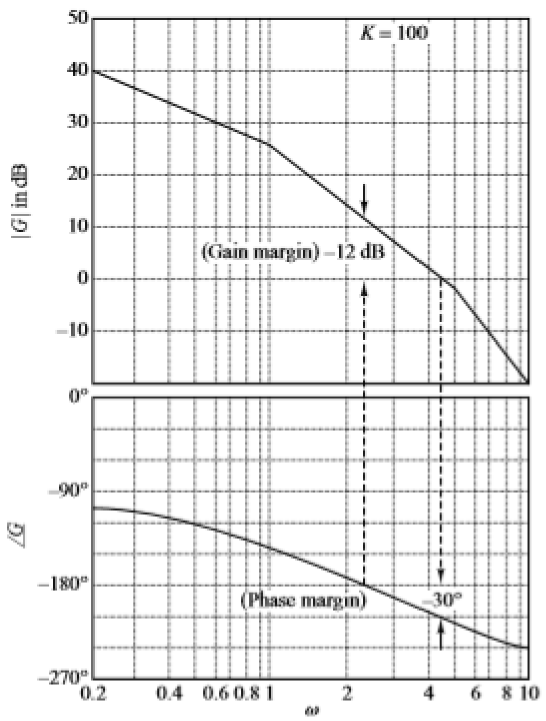


Example 1

Obtain the phase and gain margins of the system shown for the two cases where $K = 10$ and $K = 100$.







The phase and gain margins for $K = 10$ are Phase margin= $21°$, Gain margin= $8\ dB$ Therefore,the system gain may be increased by $8\ dB$ before the instability occurs. Increasing the gain from $K = 10\ to\ K = 100$ shifts the $0\ dB$ axis down by $20\ dB$,as shown.The phase and gain margins are Phase margin=$-30°$, Gain margin=$-12\ dB$ Thus,the system is stable for $K = 10$, but unstable for $K = 100$

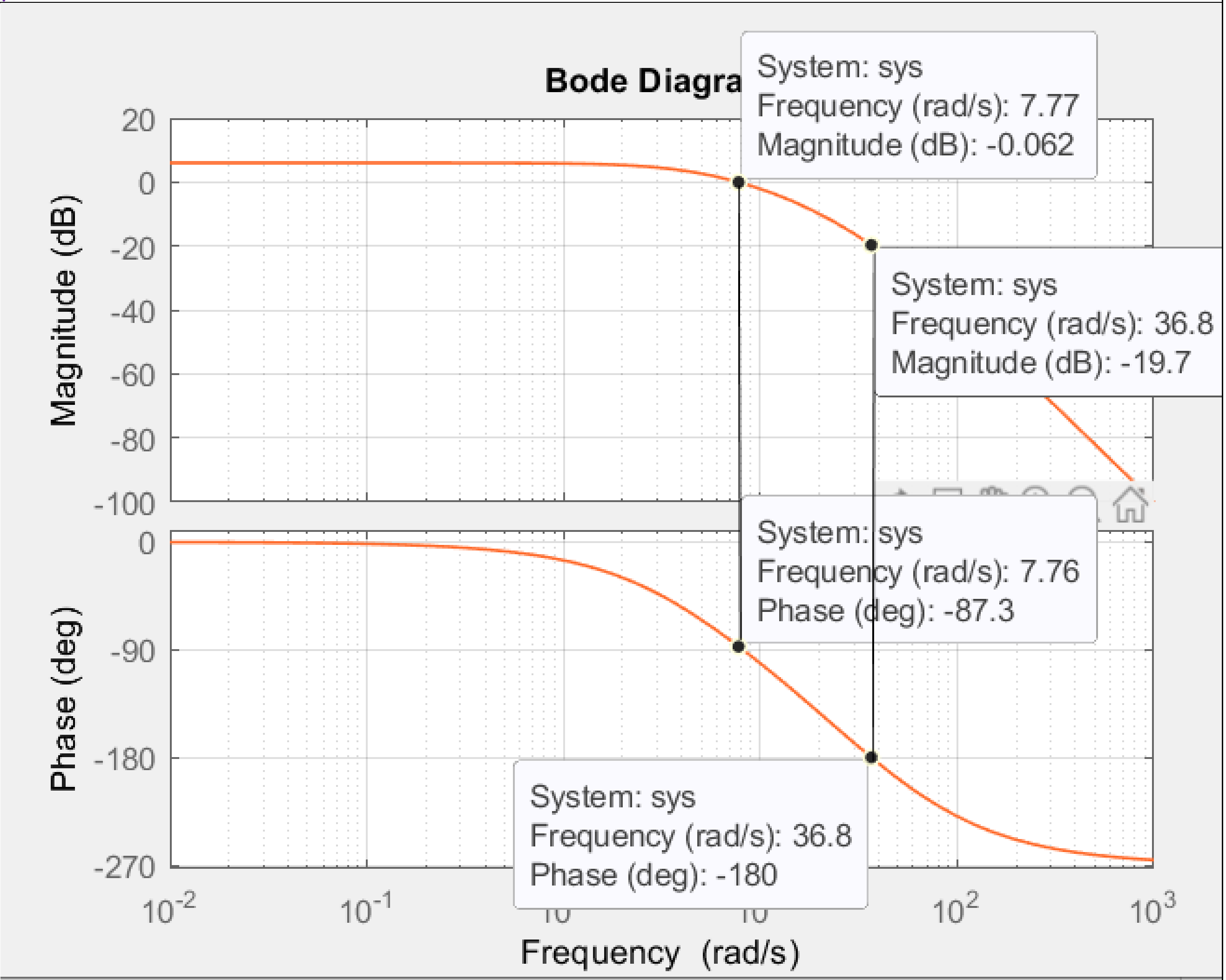
Example 2

For the system whose transfer function is $G(s) = \frac{K}{(s+5)(s+20)(s+50)}$, do the following:

a. Draw the Bode plot.

b. Find the range of K for stability from your Bode plots.

c. Evaluate gain margin, phase margin, zero dB frequency, and 180^0 frequency from your Bode plots for $K = 10000$.



gain margin of 19.7 dB.

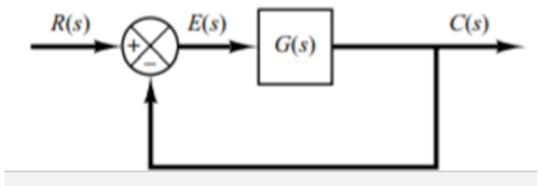
Phase margin is 92.7^0

$$20 \log k = 19.7 \qquad \ggg \qquad k = 10^{\frac{19.7}{20}} = 9.66$$

Therefore, the gain can be increased to $k \times 10000 = 96600$ before the system becomes unstable.

Relation between system type and Bode plots

Consider the unity feedback control system. The position, velocity , and acceleration error constants describe the low frequency behavior of type 0, type 1, and type 2 systems respectively.



For a given system, only one of the static error constants in finite and significant.

The type of the system determines the slope of the log-magnitude curve at low frequencies. Hence information concerning the magnitude of the steady state error of a control system to a given input can be obtained from the observation of the low frequency region of the log-magnitude plot.

Position Error Constant

Let us assume that G(s) represents a type 0 system

$$G(s) = \frac{K\,(1+T_a s)(1+T_b s)\ldots\ldots(1+T_m s)}{(1+T_1 s)(1+T_2 s)\ldots\ldots(1+T_n s)}$$

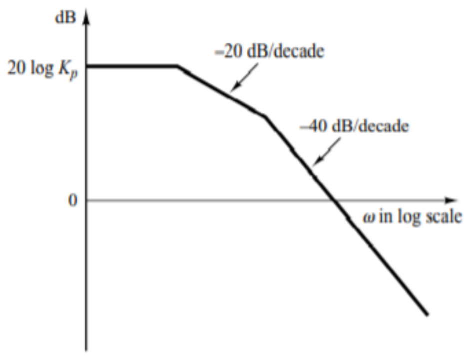
$$K_p = \lim_{s\to 0} G(s) = K$$

$$G(j\omega) = \frac{K\,(1+j\omega T_a)(1+j\omega T_b)\ldots\ldots(1+j\omega T_m)}{(1+j\omega T_1)(1+j\omega T_2)\ldots\ldots(1+j\omega T_n)}$$

$20\log|G(j\omega)| = 20\log K + 20\log|1+j\omega T_a| + 20\log|1+j\omega T_b| \ldots. +20\log|1+j\omega T_m| - 20\log|1+j\omega T_1| \ldots\ldots -20\log|1+j\omega T_n|$

At very low frequencies (of the order $\omega \to 0.01, or\ 0.001$; $20\log|G(j\omega)| \to 20\log K = 20\log K_p$

The low frequency asymptote is a horizontal line at $20\log_{10} K_p$



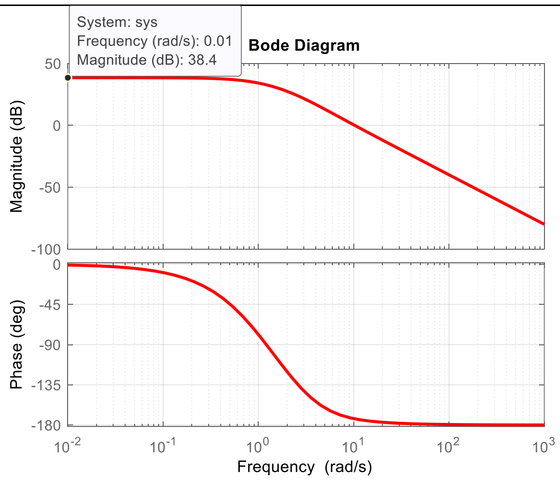
Example 3. $G(s) = \frac{100(s+5)}{(s+1)(s+2)(s+3)}$

$$K_p = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{100(s+5)}{(s+1)(s+2)(s+3)} = 83.3$$

The Bode diagram of this system is shown. From the magnitude plot, it can be seen that the log-magnitude curve is horizontal when the frequency is small. The log magnitude at low frequency is apprximately equal to $20\log_{10} K_p$.

i.e., $20\log_{10} K_p = 38.4\ dB$

$$K_p = 10^{\frac{38.4}{20}} = 83.2$$



Velocity Error Constant

Let us assume that $G(s)$ represents a type 1 system

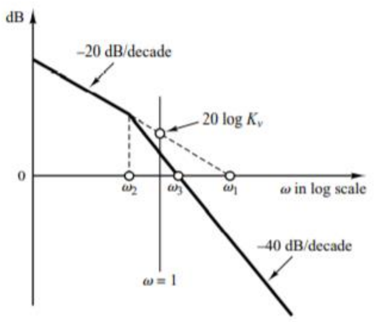
$$G(s)=\frac{K\,(1+T_a s)(1+T_b s)\ldots\ldots(1+T_m s)}{s(1+T_1 s)(1+T_2 s)\ldots\ldots(1+T_p s)}$$

$$K_v=\lim_{s\to 0} sG(s)=\lim_{s\to 0}\frac{K\,(1+T_a s)(1+T_b s)\ldots\ldots(1+T_m s)}{(1+T_1 s)(1+T_2 s)\ldots\ldots(1+T_p s)}$$

$$G(j\omega)=\frac{K\,(1+j\omega T_a)(1+j\omega T_b)\ldots\ldots(1+j\omega T_m)}{(j\omega)(1+j\omega T_1)(1+j\omega T_2)\ldots\ldots(1+j\omega T_p)}$$

The intersection of the initial $-20\ dB/decade$ line or its extension with the vertical line $\omega=1$ will give $20\log K_v$

When $\omega \ll 1$, $G(j\omega)=\frac{1}{j\omega}\lim_{j\omega\to 0}\frac{K\,(1+j\omega T_a)(1+j\omega T_b)\ldots\ldots(1+j\omega T_m)}{(1+j\omega T_1)(1+j\omega T_2)\ldots\ldots(1+j\omega T_p)}=\frac{K_v}{j\omega}$



$$20\log_{10}|G(j\omega)|=20\log_{10}\left|\frac{K_v}{j\omega}\right|=20\log_{10}K_v-20\log_{10}|j\omega|\ =20\log_{10}K_v-20\log_{10}\omega$$

When $\omega=1$; $20\log_{10}|G(j\omega)|=20\log_{10}K_v$ or $K_v=|G(j\omega)|_{\omega=1}$

Let the initial $-20\ dB/decade$ line or its extension meets the x-axis at ω_1. Then $20\log_{10}|G(j\omega)|=0$,

Hence $20\log_{10}K_v-20\log_{10}\omega_1\ =0$; or $\omega_1=K_v$

Example 4 $G(s)=\frac{10}{s(s+4)}$

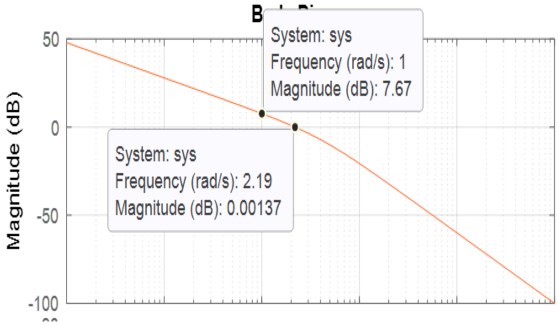
$$K_v=\lim_{s\to 0} sG(s)=\lim_{s\to 0} s\frac{10}{s(s+4)}=2.5$$

$$K_v=|G(j\omega)|_{\omega=1}=\left|\frac{10}{(j\omega)(j\omega+4)}\right|_{\omega=1}$$

$$=\left\{\frac{|10|}{|(j\omega)||(j\omega+4)|}\right\}_{\omega=1}=\frac{|10|}{|(j)||(j+4)|}=\frac{10}{1\times\sqrt{17}}=2.42$$

From the graph, $7.67\ dB=20\log_{10}K_v$

$$K_v=10^{\frac{7.67}{20}}=2.42$$



Example 5 $G(s)=\frac{10(s+5)}{s(s+4)(s+3)}$

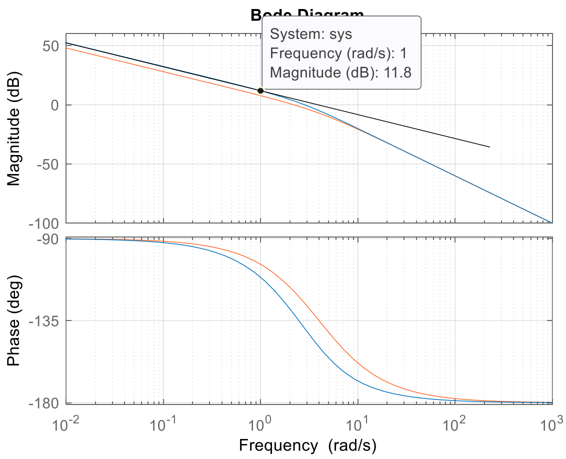
$$K_v=\lim_{s\to 0} sG(s)=\lim_{s\to 0} s\frac{10(s+5)}{s(s+4)(s+3)}=4.17$$

$$K_v=|G(j\omega)|_{\omega=1}=\left|\frac{10(j\omega+5)}{(j\omega)(j\omega+4)(j\omega+3)}\right|_{\omega=1}$$

$$=\left\{\frac{10(j\omega+5)}{(j\omega)(j\omega+4)(j\omega+3)}\right\}_{\omega=1}=\frac{|10||(j+5)|}{|(j)||(j+4)||(j+3)|}=\frac{10\sqrt{26}}{1\times\sqrt{17}\sqrt{10}}=3.9$$

From the graph, $11.8\ dB=20\log_{10}K_v$

$$K_v=10^{\frac{11.8}{20}}=3.9$$



Acceleration Error Constant

Let us assume that G(s) represents a type 2 system

$G(s) = \frac{K\,(1+T_a s)(1+T_b s)\ldots\ldots(1+T_m s)}{s^2(1+T_1 s)(1+T_2 s)\ldots\ldots(1+T_p s)}$

$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0}\frac{K\,(1+T_a s)(1+T_b s)\ldots\ldots(1+T_m s)}{(1+T_1 s)(1+T_2 s)\ldots\ldots(1+T_p s)}$

$$G(j\omega) = \frac{K\,(1+j\omega T_a)(1+j\omega T_b)\ldots\ldots(1+j\omega T_m)}{(j\omega)^2(1+j\omega T_1)(1+j\omega T_2)\ldots\ldots(1+j\omega T_p)}$$

The intersection of the initial $-40\ dB/decade$ line or its extension with the vertical line $\omega = 1$ will give $20\log K_a$

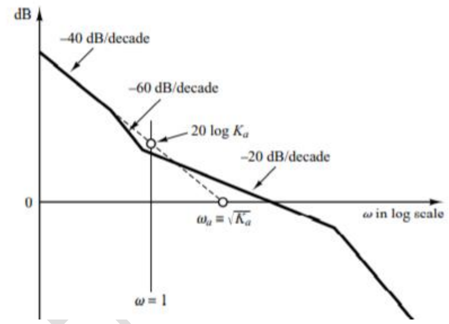
When $\omega \ll 1$, $G(j\omega) = \frac{1}{(j\omega)^2}\lim_{j\omega\to 0}\frac{K\,(1+j\omega T_a)(1+j\omega T_b)\ldots\ldots(1+j\omega T_m)}{(1+j\omega T_1)(1+j\omega T_2)\ldots\ldots(1+j\omega T_p)} = \frac{K_a}{(j\omega)^2}$

$20\log_{10}|G(j\omega)| = 20\log_{10}\left|\frac{K_a}{(j\omega)^2}\right| = 20\log_{10}K_a - 40\log_{10}|j\omega| \;= 20\log_{10}K_a - 40\log_{10}\omega$

When $\omega = 1$; $\quad 20\log_{10}|G(j\omega)| = 20\log_{10}K_a$ or $\quad K_a = |G(j\omega)|_{\omega=1}$

Let the initial $-40\ dB/decade$ line or its extension meets the x-axis at ω_a. Then $20\log_{10}|G(j\omega)| = 0$, Hence

$20\log_{10}K_a - 40\log_{10}\omega_a = 0$; or $\quad \omega_a = \sqrt{K_a}$



Example 6 $\quad G(s) = \frac{10(s+5)}{s^2(s+4)(s+3)}$

$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} s^2\frac{10(s+5)}{s^2(s+4)(s+3)} = 4.17$

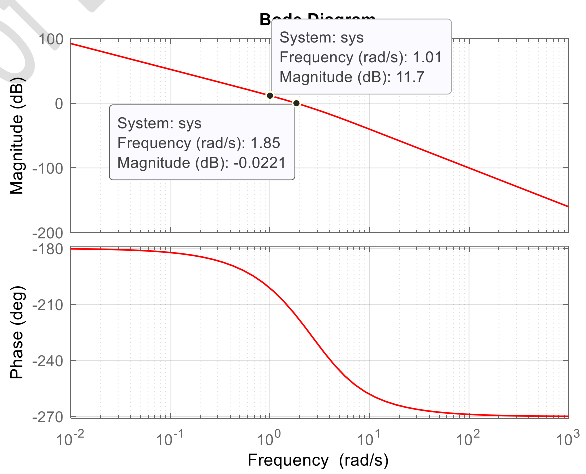
$K_a = |G(j\omega)|_{\omega=1} = \left|\frac{10(j\omega+5)}{(j\omega)^2(j\omega+4)(j\omega+3)}\right|_{\omega=1}$

$= \frac{|10||(j+5)|}{|(j)^2||(j+4)||(j+3)|} = \frac{10\sqrt{26}}{1\times\sqrt{17}\sqrt{10}} = 3.9$

From the graph, $11.7\ dB = 20\log_{10}K_a$

$$K_a = 10^{\frac{11.8}{20}} = 3.9$$

It can be also seen from the plot that log magnitude is equal to 0 dB when $\omega_a = 1.85$. Therefore, $K_a = {\omega_a}^2 = 3.4$



Correlation between step transient response and open loop frequency response

The maximum overshoot in the unit step response of a standard second order system can be exactly correlated with the resonant peak magnitude in the frequency response. Hence essentially the same information about the system dynamics is contained in the frequency response as in the transient response

Consider a unity negative feed back system with forward path transfer function $G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$. The closed loop transfer function will be $T(s) = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s^2+2\zeta\omega_n s+\omega_n^2)}$. For a unit step input, the output $c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \theta)$ where $\omega_d = \omega_n\sqrt{1-\zeta^2}$. The maximum overshoot for the unit step response is given by $M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$

$$G(j\omega) = \frac{\omega_n^2}{j\omega(j\omega + 2\zeta\omega_n)}$$

$$|G(j\omega)| = \left|\frac{\omega_n^2}{j\omega(j\omega + 2\zeta\omega_n)}\right| = \frac{|\omega_n^2|}{|j\omega||j\omega + 2\zeta\omega_n|} = \frac{\omega_n^2}{\omega\sqrt{\omega^2 + 4\zeta^2\omega_n^2}}$$

$$< G(j\omega) = 0 - 90 - \tan^{-1}\frac{\omega}{2\zeta\omega_n}$$

When $|G(j\omega)| = 1$, $\frac{\omega_n^2}{\omega\sqrt{\omega^2+4\zeta^2\omega_n^2}} = 1$ $\quad ⋙ \omega_n^2 = \omega\sqrt{\omega^2 + 4\zeta^2\omega_n^2}$ $⋙\ \omega_n^4 = \omega^2(\omega^2 + 4\zeta^2\omega_n^2)$

$$\omega^4 + 4\zeta^2\omega_n^2\omega^2 - \omega_n^4 = 0$$

$$\omega^2 = \frac{-4\zeta^2\omega_n^2 \pm \sqrt{16\zeta^4\omega_n^4 + 4\omega_n^4}}{2} = \omega_n^2\left(-2\zeta^2 \pm \sqrt{1+\zeta^4}\right)$$

$$\omega = \omega_n\sqrt{-2\zeta^2 + \sqrt{1+\zeta^4}}$$

ie., $20\log|G(j\omega)| = 0\ dB$ when $\omega = \omega_n\sqrt{-2\zeta^2 + \sqrt{1+\zeta^4}}$ and the corresponding phase angle is

$$< G(j\omega) = -90 - \tan^{-1}\frac{\omega_n\sqrt{-2\zeta^2 + \sqrt{1+\zeta^4}}}{2\zeta\omega_n}$$

The Phase margin $\gamma = 180 + < G(j\omega) = 90 - \tan^{-1}\frac{\omega_n\sqrt{-2\zeta^2+\sqrt{1+\zeta^4}}}{2\zeta\omega_n} = \tan^{-1}\frac{2\zeta}{\sqrt{-2\zeta^2+\sqrt{1+\zeta^4}}}$

The frequency ω_B at which the magnitude of the closed loop frequency response is $3\ dB$ below its zero frequency (low frequency) value is called the cut-off frequency. The closed loop system filters out signal components whose frequencies are greater than the cut-off frequency and transmits those signal components with frequencies lower than the cut-off frequency. The frequency range $0 \le \omega \le \omega_B$ in which the magnitude of the closed loop does not drop $-3\ dB$ is called the band width of the system.

The bandwidth ω_B is a system's ability to faithfully reproduce an input signal, or how well the system can track the input sinusoid. The resonant frequency ω_r and $-3\ dB$ bandwidth can be related to the speed of transient response. As the bandwidth ω_B increases, the rise time of the step response of the system will decrease. The overshoot to a step input can be related to M_p through the damping ratio ζ. The bandwidth of a system ω_B can be approximated as ω_n

$$\omega_{BW} = \omega_n\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

Example 7

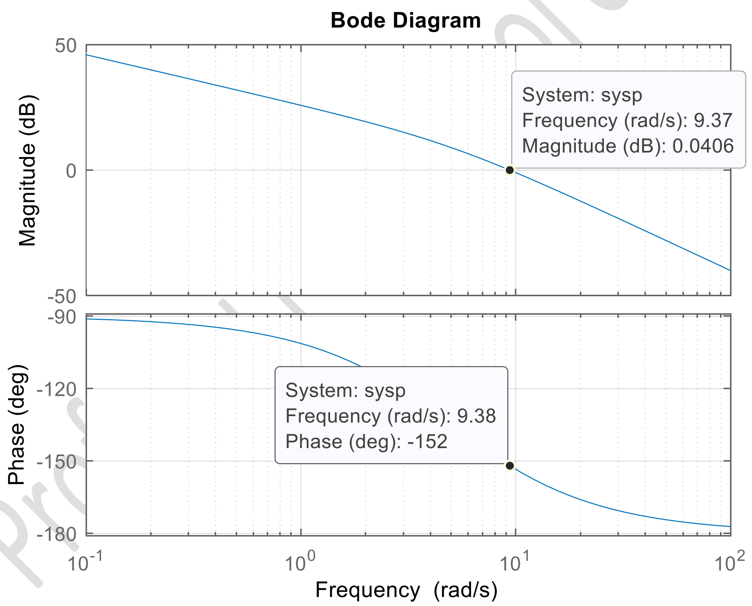
A negative unity feedback system has an open loop transfer function $G(s) = \frac{100}{s(s+5)}$, determine gain margin, phase margin, gain cross over frequency, phase cross over frquency, the velocity error constant from the open loop frequency response.

Also determine the resonant frequency, resonant magnitude and band width from the closed loop frequency response.

Open-loop frequency response

$$G(s) = \frac{100}{s(s+5)} \ggg G(j\omega) = \frac{100}{(j\omega)5\left(1+\frac{j\omega}{5}\right)} = \frac{20}{(j\omega)\left(1+\frac{j\omega}{5}\right)}$$

	20		$(j\omega)^{-1}$		$(1+j0.2\omega)^{-1}$ $T=0.2$		Σ	
ω	Mag $20\log_{10} K$	Phase $\tan^{-1} 0$	Mag $-20\log(\omega)$	Phase $\tan^{-1}\frac{-\frac{1}{\omega}}{0}$	Mag $-10\log(1+\omega^2T^2)$	Phase $\tan^{-1}(-\omega T)$	Mag dB	Phase deg
0.01	26.02	0	40	-90	0	-0.11	66	-90
0.1	26.02	0	20	-90	0	-1.1	46	-91.1
1	26.02	0	0	-90	0	-11.3	26	-101.3.
5	26.02	0	-14	-90	-3	-45	9	-135
10	26.02	0	-20	-90	-7	-63.4	6	-153.4
100	26.02	0	-40	-90	-26	-87.3	-40	-177.3
1000	26.02	0	-60	-90	-46	--89.8	-80	-179.8



gain_margin = Inf

phase_margin = 28.02 deg

gain_crossover_freq = 9.39

phase_crossover_freq= Inf

At $\omega = 1$, $20\log|G(j\omega)| = 25.8\ dB$

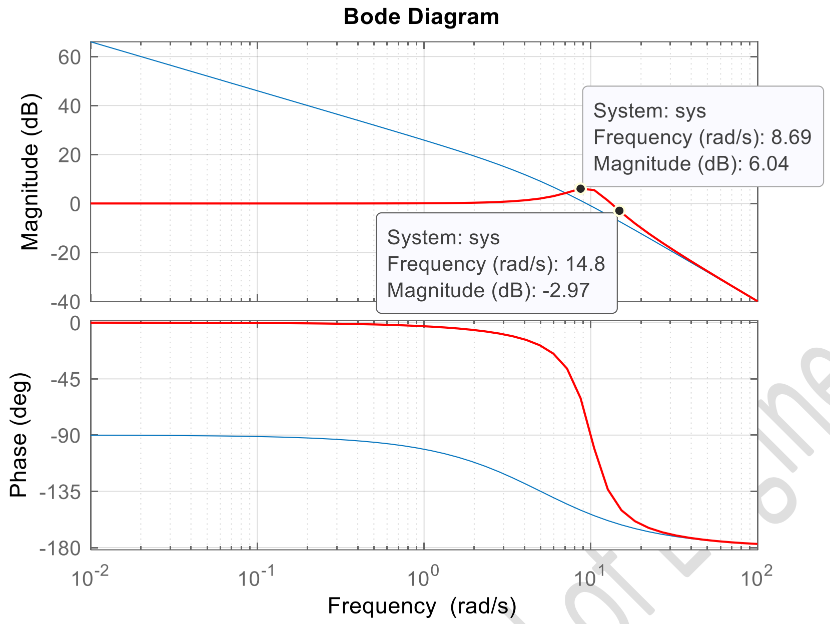
Therefore, the position error constant $K_v = 10^{\frac{25.8}{20}} = 19.5$

$$\gamma = \tan^{-1}\frac{2\zeta}{\sqrt{-2\zeta^2+\sqrt{1+\zeta^4}}} \gg \frac{2\zeta}{\sqrt{-2\zeta^2+\sqrt{1+\zeta^4}}} = 0.639 \gg \frac{4\zeta^2}{-2\zeta^2+\sqrt{1+\zeta^4}} = 0.4 \gg$$

$$4\zeta^2 = -0.8\zeta^2 + 0.4\sqrt{1+\zeta^4} \quad \ggg 144\zeta^4 = (1+\zeta^4) \ggg \zeta = 0.289$$

$$\%OS = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 38.7\%$$

The closed-loop transfer function $T(s) = \frac{1}{0.5s^3+1.5s^2+s+1} = \frac{1}{(s+2.5214)(s^2+0.4786s+0.7933)}$



resonant_peak = 6.04 dB

resonant_freq=8.69 rad/s

bandwidth=15 rad/s

Matlab code for Bode plot

```
clc
nump=[100];
denp=[1 5 0];
sysp=tf(nump,denp) % open loop transfer function
sys=feedback(sysp,1) % unity negative feedback loop
w=logspace(-3,3);    % frequency range from 0.001 to 1000
bode(sysp,w)         % open loop frequency response

[Gm,pm, wcp, wcg]=margin(sysp);
GmdB=20*log10(Gm);
gain_margin= GmdB
phase_margin=pm
gain_crossover_freq=wcg
phase_crossover_freq=wcp
hold on
bode(sys,w)          % closed loop frequency response
[mag,phase,w]=bode(sys,w);

[Mp,k]=max(mag);
resonant_peak=20*log10(Mp)
resonant_freq=w(k)
 n=1;
while (20*log(mag(n)))>-3;  % checking log magnitude is > -3dB

 n=n+1;

end
bandwidth=w(n)
mag_dB=20*log10(mag(n))
```

References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons